

Twofold monoidal structures

Malthe Sporring

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Graduate School of Mathematics

University of Edinburgh

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Abstract

We discuss two-fold symmetric monoidal ∞ -categories à la Barwick. These are, roughly speaking, ∞ -categories with two symmetric monoidal structures with the same unit where one monoidal product distributes over the other in the sense of Eckmann-Hilton, but laxly. Using a version of the straightening-unstraightening equivalence, they can be modeled as certain fibrations over $\mathbb{E}_*$, the category of pointed cographs. Alternatively, they are expected to correspond to symmetric duoidal ∞ -categories, in the sense of Torii [Tor24], where the two monoidal structures have the same unit.

By a result of Haugseng and Kock [HK22], the category of ∞ -operads form a full subcategory of the category of symmetric monoidal ∞ -categories equipped with a symmetric monoidal functor to $\mathbb{F}$, the category of finite sets with monoidal product given by disjoint union. We propose this theorem may be extended to give a definition of "twofold operads", giving an ∞ -categorical generalisation of the compound pseudo-tensor categories of Beilinson and Drinfeld [BD04]. We describe partial progress to making this definition explicit, which involves a description of $\mathbb{H}$, the free two-fold symmetric monoidal category on an object equipped with a monoid structure for the left product and a comonoid structure for the right product. Finally, we discuss potential applications of this theory.

Contents

Abstract	2
1 Literature review	4
1.1 Introduction	4
1.1.1 Conventions	5
1.2 Operator categories and their operads	5
1.2.1 Operator categories	6
1.3 Operads from monoidal categories	8
2 Twofold monoidal structures	12
2.1 Cographs	12
2.1.1 Coloured cographs	14
2.2 Twofold monoidal categories	15
2.3 The category $\mathbb{H}$	16
2.4 Next steps	19
Bibliography	19

Chapter 1

Literature review

1.1 Introduction

∞ -operads are combinatorial gadgets for controlling homotopy-coherent algebraic theories. For example, there exist ∞ -operads E_n whose algebras are exactly E_n -algebras. Lurie defines (symmetric) operads as certain inner fibrations over $\mathbb{F}_*$, the category of pointed finite sets [Lur17], which should be thought of as a generalisation of the category of operators associated to a 1-operad. Lurie's definition is known to be equivalent other models of ∞ -operads, namely the dendroidal ∞ -operads of Moerdijk-Weiss [MW07] [HHM15] [HM24], and the complete Segal ∞ -operads of Barwick [Bar18] [CHH18].

Some algebras may be better described using a different formulation entirely. For example, the associative operad *Assoc* is more easily described as the terminal **non-symmetric** operad, i.e. as a certain fibration over Δ^{op} . Barwick's notion of a perfect operator category [Bar18] provides sufficient conditions for a category ϕ to admit a sensible notion of ϕ -operads. This is further generalised by the algebraic patterns of Chu and Haugseng [CH21]. Noting that symmetric monoidal ∞ -categories are a special case of ∞ -operads, these formulations also generalise the notion of symmetric monoidal category.

A structure not captured by either notion is that of a duoidal category. This is, roughly, a category C equipped with two symmetric tensor products $(\otimes_A, 1_A)$, $(\otimes_B, 1_B)$ such that one distributes laxly over the other. The 1-categorical version is due to Aguiar and Mahajan [AM10], under the name of 2-monoidal category, and an ∞ -categorical generalisation has been developed by [Tor24] in the form of certain mixed fibrations to Δ^{op} . When the two units for the tensor products coincide, we call the category twofold-monoidal. Twofold monoidal 1-categories have been studied by [BFSV98] and [BD04], in the latter under the name of compound tensor category. An ∞ -categorical version is proposed by Barwick [Bar24], as certain fibrations over $\mathbb{E}_*$, the category of pointed reflexive cographs (see Section 2.1).

We are interested in describing an operadic version of this concept, giving an ∞ -categorical version of compound pseudo-tensor categories in the sense of [BD04].

The paper is organised as follows. Chapter 1 is devoted to a literature review.

In Section 1.2 we will review the notion of an operator category ϕ and the related notion of ϕ –operad. Section 1.3 is devoted to the symmetric monoidal envelope, and the proof by Haugseng and Kock [HK22] that operads form a full subcategory of the category of symmetric monoidal functors equipped with a symmetric monoidal functor to $\mathbb{F}^{\sqcup}$. Our purpose of reviewing this result is to speculate on the existence of a twofold monoidal envelope comonad on the category of twofold monoidal categories, from which we can *define* twofold operads as certain twofold monoidal categories with a twofold monoidal functor to $\mathbb{E}$.

Chapter 2 is devoted to twofold monoidal structures. In Section 2.1 we describe cographs and the categories $\mathbb{E}$ and $\mathbb{D}$, and in Section 2.2, review Barwick’s definition of a twofold monoidal category. $\mathbb{D}$ and $\mathbb{E}$ are free twofold monoidal on a monoid for, respectively, the left and the right product. In Section 2.3 we introduce the category $\mathbb{H}$, which is the free twofold monoidal category on a monoid for the left product that is a comonoid for the right product. We speculate on a definition of twofold operads described as a fibration over $\Lambda(\mathbb{H})$. Finally, in Section 2.4 we discuss open questions and potential next steps for the project.

1.1.1 Conventions

- We use implicit ∞ -conventions. A category will always refer to an $(\infty, 1)$ -category, an operad to an $(\infty, 1)$ –operad, and so on, unless explicitly specified.
- An operad will always refer to the coloured or multicategory case, unless explicitly specified. This is, broadly speaking, in contrast to the language used in the 1-categorical literature, but consistent with the language in the ∞ –categorical literature.
- A symmetric monoidal functor will mean *strong* symmetric monoidal functor, i.e. an edge of $Cat_{\infty/\mathbb{F}_*}$ between symmetric monoidal categories that preserves coCartesian edges. Since we only consider commutative structures in this paper, we will sometimes leave "symmetric" implicit and so talk about monoidal functors of monoidal categories.

1.2 Operator categories and their operads

Let’s fix some notation. Let $\mathbb{F}$ be the category of finite sets and functions, $\mathbb{F}_*$ the category of pointed finite sets, $\mathbb{O}$ be the category of ordered finite sets and order-preserving maps, and Δ the category of non-empty ordered finite sets. Let $\langle n \rangle = \{*, 1, \dots, n\} \in F_*$, $\langle n \rangle^\circ = \{1, \dots, n\} \in F$, $[n] = (0, 1, \dots, n) \in \Delta$, $[n]^\circ = (1, \dots, n) \in O$.

Definition 1.2.1. We call a morphism of pointed sets $f : \langle n \rangle \rightarrow \langle m \rangle$ **inert** if the induced function $f : \langle n \rangle \setminus f^{-1}(*) \rightarrow \langle m \rangle^\circ$ is a bijection, and call f **active** if $f^{-1}(*) = \{*\}$.

The collection of inert and active morphisms in $\mathbb{F}_*$ forms an (orthogonal) factorisation system on $\mathbb{F}_*$ in the sense of [FK72]. Now recall the following definition due to Lurie.

Definition 1.2.2. An ∞ -**operad** is an inner fibration $p : \mathcal{O}^\otimes \rightarrow \mathbb{F}_*$ such that

1. Inert morphisms in $\mathbb{F}_*$ admit p -coCartesian lifts.
2. For each $f : \langle m \rangle \rightarrow \langle n \rangle$ and p -coCart mors $C' \rightarrow C'_i$ lying over $\rho^i : \langle n \rangle \rightarrow \langle 1 \rangle$ for $1 \leq i \leq n$, the induced map

$$\mathrm{Map}_{\mathcal{O}^\otimes}^f(C, C') \rightarrow \prod_i \mathrm{Map}_{\mathcal{O}^\otimes}^{\rho^i \circ f}(C, C'_i)$$

is a homotopy equivalence.

3. The p -coCartesian lifts of the inert maps $\rho^i : \langle n \rangle \rightarrow \langle 1 \rangle$ assemble to an equivalence $\mathcal{O}_{\langle n \rangle}^\otimes \simeq \prod_i \mathcal{O}_{\langle i \rangle}$.

Remark 1.2.3. Let us record the properties of $\mathbb{F}_*$ we used to make this definition.

- We used the inert-active factorisation system on $\mathbb{F}_*$. Notice F_* is the category of free algebras over the monad T on $\mathbb{F}$ that freely adds a point. The inert-active factorisation system on $\mathbb{F}_*$ can be described entirely using this monad.
- We singled out a special object $\langle 1 \rangle$, with respect to which we gave our Segal condition. In the operator categories story this choice will be unique, but see [CH21] for an exploration of what happens when this collection of "elementary objects" is varied.
- We used the fact that $\mathbb{F}_*$ has finite products, and that there are only finitely many inert morphisms $\rho^i : \langle n \rangle \rightarrow \langle 1 \rangle$, so the Segal condition makes sense. In the operator categories story the ρ^i come from a notion of point $i \in \langle n \rangle$ together with the existence of classifying morphisms $\rho^i : \langle n \rangle \rightarrow \langle 1 \rangle$ whose fiber over $1 \in \langle 1 \rangle$ is i .
- Given a symmetric monoidal category, i.e. a coCartesian fibration $C^\otimes \rightarrow \mathbb{F}_*$, we write $C := C_{\langle 1 \rangle}^\otimes$ for the underlying category.

To generalize Lurie's definition, we really want to study F and its canonical monad T .

1.2.1 Operator categories

This section is based on [Bar18], where the central definition is the following.

Definition 1.2.4. An **operator category** is an essentially small category ϕ with finite Hom sets, a terminal object $*$, and fibers.

For $I \in \phi$ we define the set of points $|I|$ to be the set $\phi(*, I)$.

Example 1.2.5. The categories $\mathbb{F}$ and $\mathbb{O}$ are operator categories.

Definition 1.2.6. A universal point is a pair (T, o) with $o : * \rightarrow T$ with the property that for every $I \in \phi$ and $i \in |I|$ there exists a characteristic map $\chi_i : I \rightarrow T$ whose fiber over o is i .

Example 1.2.7. $(\langle 1 \rangle, 1)$ is a universal point in $\mathbb{F}$. $((\perp < 0 < \top), 0)$ is a universal point in $\mathbb{O}$.

Definition 1.2.8. An operator category is perfect if it has a universal point and the functor $Fib : \phi_{/T} \rightarrow \phi$ assigning $I \rightarrow T$ to the fiber I_o has a right adjoint.

Let ϕ be a perfect operator category and E be the right adjoint of Fib . Let $U : \phi_{/T} \rightarrow \phi$ be the forgetful functor and $T := U \circ E$. As Barwick proves, T has the structure of a monad on ϕ [Bar18]. Let us describe its unit. First note that Fib has a left adjoint p where $p(C) = (C \rightarrow * \xrightarrow{o} T)$. The adjunctions (Fib, E) and (p, Fib) exhibit Fib as both a localisation and colocalisation, that is, we have natural isomorphisms $\eta : id \rightarrow Fib \circ p$, $\epsilon : Fib \circ E \rightarrow id$. We then get a natural transformation $p \rightarrow E \circ fib \circ p \rightarrow E$ from the unit of (Fib, E) , which composed with the forgetful functor gives a natural transformation $id \cong U \circ p \rightarrow U \circ E := T$, which we define as the unit of T . We appeal to [Bar18] for a description of the multiplication.

Definition 1.2.9. The Leinster category $\Lambda(\phi)$ is the Kleisli category of T . That is, it has the same objects as ϕ and $\Lambda(\phi)(I, J) = \phi(I, T(J))$. Composition is given by

$$I \xrightarrow{f} TJ \xrightarrow{Tg} T^2K \xrightarrow{\mu} TK.$$

Example 1.2.10. $\mathbb{F}$ is perfect, and its Leinster category is the category of pointed finite sets $\mathbb{F}_*$. $\mathbb{O}$ is perfect, and its Leinster category is Δ^{op} .

Definition 1.2.11. A morphism $I \rightarrow J$ in $\Lambda(\phi)$ is called **inert** if the corresponding morphism $J \times_{TJ} J \rightarrow I$ is an iso. It is called **active** if it factors as $I \rightarrow TI \xrightarrow{Tf} TJ$. The inert and active morphisms form an orthogonal factorisation system on $\Lambda(\phi)$.

Example 1.2.12. In the case of $\phi = \mathbb{F}$ the induced factorisation system on $\mathbb{F}_*$ is exactly the one given by Definition 1.2.1.

The data of an operator category and its canonical monad T is sufficient to construct a Lurie-style theory of operads. For a perfect operator category ϕ , $I \in \phi$ and $i \in |I|$, denote by ρ_i the inert morphism $I \rightarrow *$ in $\Lambda(\phi)$ corresponding to $\chi_i : I \rightarrow T$ in ϕ .

Definition 1.2.13. Let ϕ be a perfect operator category. A ϕ -**operad** is an inner fibration $p : \mathcal{O}^\otimes \rightarrow \Lambda(\phi)$ such that

1. Inert morphisms in $\Lambda(\phi)$ admit p -coCartesian lifts.

2. For each $f : I \rightarrow J$ in $\Lambda(\phi)$ and p -coCart mors $C' \rightarrow C'_i$ lying over $\rho^i : J \rightarrow \{i\}$ for $i \in |J|$ the induced map

$$\mathrm{Map}_{O^\otimes}^f(C, C') \rightarrow \prod_i \mathrm{Map}_{O^\otimes}^{\rho^i \circ f}(C, C'_i)$$

is a homotopy equivalence.

3. For every $J \in \phi$, the p -coCartesian edges lying over the inert morphisms $\rho_i : J \rightarrow \{i\}$ assemble to an equivalence

$$O_J^\otimes \simeq \prod_{i \in |J|} O_{\{i\}}^\otimes.$$

Remark 1.2.14. Barwick also gives a second definition of operad over an operator category which does not require that the operator category is perfect. These are the so-called complete Segal operads. In the case that ϕ is perfect, the two definitions are equivalent. [Bar18]

1.3 Operads from monoidal categories

The purpose of this section is to summarize results of Haugseng-Kock which allow one to describe the operads (i.e. $\mathbb{F}$ -operads) as certain symmetric monoidal categories equipped with a monoidal functor to $\mathbb{F}$. We will describe the theory for (symmetric) operads, i.e. $\mathbb{F}$ -operads [HK22]. The theory can also be extended to a large class of algebraic patterns¹ [BHS22].

The category of operads Op is the subcategory of $Cat_{\infty/\mathbb{F}_*}$ spanned by operads and morphisms between them that preserve cocartesian lifts of inerts. Recall Lurie's definition of a symmetric monoidal category as a coCartesian fibration $p : C^\otimes \rightarrow \mathbb{F}_*$, and that a symmetric monoidal functor is a morphism in $Cat_{\infty/\mathbb{F}_*}$ between symmetric monoidal categories that preserves all coCartesian edges. In this formulation, a symmetric monoidal category is in particular operad, and a monoidal functor is in particular an operad morphism, giving a faithful functor $U : SMonCat \hookrightarrow Op$. This functor has a left adjoint, given by the **symmetric monoidal envelope**. Informally, it takes an operad $O^\otimes$ to the category $O_{act}^\otimes$ of active morphisms in $O^\otimes$ with the symmetric monoidal product given by concatenation, which is sensible due to conditions 1 and 2 of Definition 1.2.2. Here, an active morphism in $O^\otimes$ means a morphism lying over an active morphism in $\mathbb{F}_*$.

To make the construction precise, define $Act(\mathbb{F}_*)$ as the full subcategory of $Ar(\mathbb{F}_*) := Fun(\Delta^1, \mathbb{F}_*)$ whose objects are active morphisms, and let $s, t : Act(\mathbb{F}_*) \rightarrow \mathbb{F}_*$ denote the source and target morphisms, respectively. Define

¹Specifically those that are *sound* and *extendable*

the envelope of $\pi : \mathcal{O}^\otimes \rightarrow \mathbb{F}_*$ to be the pullback

$$\begin{array}{ccc} Env(\mathcal{O})^\otimes & \longrightarrow & \mathcal{O}^\otimes \\ \downarrow & \lrcorner & \downarrow \pi \\ Act(\mathbb{F}_*) & \xrightarrow{s} & \mathbb{F}_* \end{array}$$

equipped with the functor to $\mathbb{F}_*$ induced by t . To see that underlying category of $Env(\mathcal{O})^\otimes$ is $\mathcal{O}_{act}^\otimes$, note that there is a unique active map from any object in $\mathbb{F}_*$ to $\langle 1 \rangle$, so an object of $Env(\mathcal{O})_{\langle 1 \rangle}^\otimes$ is just an object X of $\mathcal{O}^\otimes$ (with the canonical active map $!a : \pi(\mathcal{O}^\otimes) \rightarrow \langle 1 \rangle$), and a morphism from X to Y is a morphism $f : X \rightarrow Y$ and a commutative triangle

$$\begin{array}{ccc} \pi(X) & \xrightarrow{\pi(f)} & \pi(Y) \\ & \searrow !a & \swarrow !a \\ & \langle 1 \rangle & \end{array}$$

which implies $\pi(f)$ is active.

We note this construction gives a functor $Env : Op \rightarrow Cat_{\infty/\mathbb{F}_*}$. By [Lur17][2.2.4] this factors through the subcategory $SMonCat$ and is a left adjoint to the forgetful functor $U : Op \rightarrow SMonCat$, which we record as the following theorem.

Theorem 1.3.1 ([Lur17] 2.2.4.4 and 2.2.4.9). *The construction of the envelope gives a functor $Env : Op \rightarrow SMonCat$ left adjoint to $U : SMonCat \rightarrow Op$.*

Remark 1.3.2. The cocartesian lifts of $\pi : Env(\mathcal{O})^\otimes \rightarrow \mathbb{F}_*$ are identified as follows: let $\pi : \mathcal{O}^\otimes \rightarrow \mathbb{F}_*$ be an operad, let $\phi : \langle n \rangle \rightarrow \langle m \rangle$ be a morphism in $\mathbb{F}_*$ and let $(X, \alpha : \pi(X) \rightarrow \langle n \rangle) \in Env(\mathcal{O}^\otimes)$ be a lift of $\langle n \rangle$. Remark that $\phi \circ \alpha$ has a unique factorisation as an inert morphism $i : \pi(X) \rightarrow K$ followed by an active morphism $a : K \rightarrow \langle m \rangle$, as in the following diagram

$$\begin{array}{ccc} \pi(X) & \xrightarrow{i} & K \\ \alpha \downarrow & & \downarrow a \\ \langle n \rangle & \xrightarrow{\phi} & \langle m \rangle \end{array}$$

Let $\phi' : X \rightarrow K'$ be the π -coCartesian edge lifting i . Then ϕ' together with the above commutative square is the t -cocartesian lift of ϕ . Let us unpack a particular case of the t -coCartesian condition. Suppose we are given another morphism in $Env(\mathcal{O})^\otimes$ out of X as follows

$$X \xrightarrow{\psi} Y$$

$$\begin{array}{ccc} \pi(X) & \xrightarrow{\pi(\psi)} & \pi(Y) \\ \downarrow \alpha' & & \downarrow \alpha'' \\ \langle n \rangle & \xrightarrow{g} & \langle k \rangle \end{array}$$

as well as a morphism $h : \langle m \rangle \rightarrow \langle k \rangle$ such that $g = h \circ f$. Then since (inert, active) is an orthogonal factorisation system, there is a unique lift θ of the following square

$$\begin{array}{ccc} \pi(X) & \xrightarrow{\pi(\psi)} & \pi(Y) \\ \downarrow i & \nearrow \theta & \downarrow \alpha'' \\ K & \xrightarrow{g \circ a} & \langle k \rangle \end{array}$$

Hence since ϕ' is π -coCartesian, there is a lift $\theta' : K' \rightarrow Y$ such that $\theta \circ \phi \simeq \psi$. The morphism θ' together with the commutative square

$$\begin{array}{ccc} K & \xrightarrow{\theta} & \pi(Y) \\ \downarrow a & & \downarrow \alpha'' \\ \langle m \rangle & \xrightarrow{h} & \langle k \rangle \end{array}$$

is then the required lift of h .

Example 1.3.3. For the terminal operad $id : \mathbb{F}_* \rightarrow \mathbb{F}_*$ we have

$Env(\mathbb{F}_*) \equiv Act(\mathbb{F}_*)$. We can identify its underlying category with $\mathbb{F} \simeq Act(\mathbb{F}_*)_{\langle 1 \rangle}$ and notice that the monoidal product is given by disjoint union of finite sets.

Example 1.3.3 demonstrates that Env can be upgraded to a functor $Env' : Op \rightarrow SMonCat_{/\mathbb{F}^\sqcup}$. On the other hand, given a monoidal functor $C^\otimes \rightarrow \mathbb{F}^\sqcup$, one can apply the forgetful functor to Op then pull it back along the unique monoidal functor $\mathbb{F}_* \rightarrow \mathbb{F}^\sqcup$ in Op to get an operad $U'(C^\otimes) \rightarrow \mathbb{F}_*$. This construction assembles to a functor $U' : SMonCat_{/\mathbb{F}^\sqcup} \rightarrow Op$.

Theorem 1.3.4 ([HK22]). *The functor Env' is left adjoint to U' . Furthermore, Env' is fully faithful.*

Proof. The adjunction (Env', U') is a special case of [Lur08][5.2.5.1]. The unit of $Env \dashv U$ has components given by i_O in the following diagram, where every square is a pull-back square. It is well-defined because $id_{\mathbb{F}_*} \simeq s \circ id_-$.

$$\begin{array}{ccccc} \mathcal{O}^\otimes & \xrightarrow{i_O} & Env(\mathcal{O})^\otimes & \longrightarrow & \mathcal{O}^\otimes \\ \downarrow & \lrcorner & \downarrow & & \downarrow \\ \mathbb{F}_* & \xrightarrow{id_-} & Act(\mathbb{F}_*) & \xrightarrow{s} & \mathbb{F}_* \end{array}$$

Continue
from
here.

By the proof of [Lur08][5.2.5.1], the induced unit of $Env' \dashv U'$ has components given by the induced arrow in the following diagram

$$\begin{array}{ccc}
 \mathcal{O}^\otimes & \xrightarrow{p} & \mathbb{F}_* \\
 \text{---} \swarrow \text{---} & & \downarrow \\
 & \mathbb{F}_* \times_{Act(\mathbb{F}_*)} Env(\mathcal{O})^\otimes & \longrightarrow \mathbb{F}_* \\
 \text{---} \searrow \text{---} & \downarrow & \downarrow \\
 & Env(\mathcal{O})^\otimes & \longrightarrow Act(\mathbb{F}_*)
 \end{array}$$

i_O (curved arrow from $\mathcal{O}^\otimes$ to $Env(\mathcal{O})^\otimes$)

which by what we have just shown is an equivalence. We conclude by recalling a left adjoint is fully faithful if and only if its unit is an equivalence. $\square$

Our hope is to use Theorem 1.3.4 as a definition for twofold operads, that is, to identify the comonad on the category of twofold symmetric monoidal categories corresponding to $Env \circ U$. Alternatively, Haugseng and Kock also give an explicit characterization of the subcategory of $SMonCat_{/\mathbb{F}^\sqcup}$ cut out by Env' in [HK22][2.4.6] which we may attempt to replicate.

Chapter 2

Twofold monoidal structures

2.1 Cographs

In this section we will describe the category of reflexive cographs and use it to give a definition of twofold symmetric monoidal category. For us, a graph is a pair (V, E) where V is a finite set of vertices, and $E \subset V \times V$ is a set of edges between vertices satisfying the property

$$(x, y) \in E \implies (y, x) \in E, \forall x, y \in V.$$

In particular, we only consider graphs with at most one edge between vertices. In other words, a graph is the same thing as a finite symmetric relation. If a, b are vertices connected by an edge we will write $a - b$. Otherwise, we will write $a \not- b$. A graph morphism is a function of vertices that preserves edges. The category of graphs and graph morphisms will be denoted *Graph*.

Definition 2.1.1. Let G be a graph. An **induced subgraph** of G is a graph H and a graph morphism $i : H \hookrightarrow G$ that is injective on vertices and bijective on edges onto its image. We say G is **free from** H if H cannot be given the structure of an induced subgraph of G .

Definition 2.1.2. Given a graph $G = (V, E)$, its dual graph $G^\vee$ is the pair $(V, V \times V \setminus E)$.

Definition 2.1.3. Denote by P_4 the graph $a - b - c - d$.

Definition 2.1.4. A **cograph** is a graph free from P_4 . A **reflexive cograph** is a cograph whose underlying relation is reflexive, and an **irreflexive cograph** is a cograph whose underlying relation is irreflexive. We denote by $\mathbb{G}$ the full subcategory of cographs, $\mathbb{E}$ the full subcategory of reflexive cographs, and $\mathbb{D}$ the full subcategory of irreflexive cographs.

Remark 2.1.5. By definition, any induced subgraph of a cograph is itself a cograph.

There are two monoidal products on *Graph*: the disjoint union of graphs $\underline{\oplus}$ given by

$$(V, E) \underline{\oplus} (V', E') = (V \sqcup V', E \sqcup E')$$

and the join $\overline{\oplus}$ given by

$$(V, E)\overline{\oplus}(V', E') = (V \sqcup V', E \sqcup E' \sqcup (V \times V') \sqcup (V' \times V)),$$

where all inclusions are canonical under the isomorphism

$$(V \sqcup V') \times (V \sqcup V') \cong (V \times V) \sqcup (V' \times V') \sqcup (V \times V') \sqcup (V' \times V).$$

These monoidal products have the same unit, the empty graph $\emptyset$. $\mathbb{G}$, $\mathbb{E}$, and $\mathbb{D}$ are all closed under $\underline{\oplus}$ and $\overline{\oplus}$.

Lemma 2.1.6. *The dual of a cograph is a cograph.*

Proof. Suppose for sake of contradiction G is a cograph and $P_4 \hookrightarrow G^\vee$ is an induced subgraph. In other words, we have $a, b, c, d \in G$ are such that $a \not\sim b, b \not\sim c, c \not\sim d$ but $a \sim c, b \sim d, a \sim d$. This implies $c - a - d - b$ is an induced subgraph of G , which is a contradiction. $\square$

Definition 2.1.7. We say a graph is **disconnected** if it can be written as a disjoint union of two non-empty graphs. A graph is **connected** if it is not disconnected. We say a graph is **totally disconnected** if $a \sim b \implies a = b$. It is **complete** if $a \neq b \implies a \sim b$.

Lemma 2.1.8. *A cograph $G \in \mathbb{G}$, where $|G| > 1$, is connected if and only if $G^\vee$ is disconnected.*

Proof. The if direction is trivial. We will prove the only if direction by induction on the size of G . The statement is clear when $|G| = 2$. Suppose the statement holds for graphs of size k , but there exists a connected graph G of size $k+1$ whose complement is also connected. Let $x \in G$ and consider the subgraph $G \setminus \{x\}$. By assumption either this graph or its dual is disconnected, so WLOG assume $G \setminus \{x\}$ is disconnected. We remark that a cograph is connected if and only if for every pair x, y of vertices there exists a vertex z such that $x \sim z \sim y$. Now choose $y \in G \setminus \{x\}$. Let z be in a different connected component to y . It follows that $y \sim x \sim z$ in G . Now let $w \sim y$. It follows by the cograph condition that $w \sim x$. Repeating this argument shows every vertex in the connected component of y is related to x . But then $\{x\}$ is a disconnected component in $G^\vee$, which is a contradiction. $\square$

We combine above lemmas to show the following.

Proposition 2.1.9. *$\mathbb{G}$ is the smallest subcategory of Graph containing the two 1-vertex graphs, and closed under finite disjoint unions and joins. $\mathbb{E}$ is the smallest subcategory of Graph containing the reflexive 1-vertex graph and closed under finite disjoint unions and joins. $\mathbb{D}$ is the smallest subcategory of Graph containing the irreflexive 1-vertex graph and closed under finite disjoint unions and joins.*

Proposition 2.1.9 is the main motivation for considering cographs. It shows, in particular, that the objects of $\mathbb{E}$ are the finite expressions that can be written using one variable and two products with the same unit.

$\mathbb{E}$, $\mathbb{G}$, and $\mathbb{D}$ are twofold monoidal categories in the sense of [BFSV98]. We also notice that the functor $\tau : \mathbb{F} \rightarrow \mathbb{E}$ which sends a set X to the complete graph on $|X|$ vertices is fully faithful, and monoidal with respect to the product $\sqcup$ on $\mathbb{F}$ and $\oplus$ on $\mathbb{E}$. Similarly, the functor $\delta : \mathbb{F} \rightarrow \mathbb{D}$ which sends a set X to the totally disconnected graph on $|X|$ vertices is fully faithful and monoidal with respect to $\underline{\oplus}$ on $\mathbb{D}$. The following proposition follows essentially by inspection, a full proof will appear in future work. We furthermore claim that the universal property of $\mathbb{D}$ and $\mathbb{E}$ can be extended to twofold monoidal ∞ -categories, as defined in Section 2.2

Proposition 2.1.10. *$\mathbb{E}$ is the free twofold monoidal category on a monoid for the left product. $\mathbb{D}$ is the free twofold monoidal category on a monoid for the right product.*

Remark 2.1.11. $\mathbb{E}$ is equivalently the free twofold monoidal category on a monoid for both products, compatibly with the interchange map. This is because a monoid X for the right product induces a monoid on the left product via the interchange map $X \underline{\oplus} X \rightarrow X \oplus X$. On the other hand, a compatible monoid for both products is uniquely determined by the monoid for the right product.

2.1.1 Coloured cographs

We can generalize the above construction to handle situations where there are more than two tensor products. We fix a poset P , whose objects we call colours, and make the following definition.

Definition 2.1.12. A P -coloured graph is a complete graph whose edges are labeled by objects of P . We call the objects of P colours. A P -coloured graph morphism is a graph morphism that preserves the poset structure on colours. Denote the category of finite P -graphs and P -morphisms by $Graph_P$. An n -coloured graph is any P -coloured graph where $|P| = n$. Letting G be a P -coloured graph, $x \in P$ and $a, b \in G$, we write $a -_x b$ if a and b are connected by an x -coloured edge.

Remark 2.1.13. A graph is exactly a coloured graph over the poset $0 < 1$.

Any category of n -coloured graphs has n symmetric monoidal products given by coloured joins, with each monoidal unit given by the empty graph. For every poset P and object $x \in P$ we have a functor $p_x : Graph_P \rightarrow Graph$ which sends a coloured graph G to the graph with the same vertices as G and where $x - y$ iff $x -_x y$. We say a graph G is x -connected if $p_x G$ is connected, and x -disconnected if $p_x G$ is disconnected.

Definition 2.1.14. Let the category of P -cographs $\mathbb{G}_P$ be the smallest subcategory of $Graph_P$ containing every 1-vertex graph and closed under finite coloured joins. Call a P -graph a P -cograph if it lies in this subcategory.

Proposition 2.1.15. *A P -coloured graph G is a P -cograph if and only if $p_x(G)$ is a cograph for every $x \in P$, and G is free from any graph with three vertices and three different coloured edges.*

Proof. We may assume there are at least two colours. The only if direction is clear. For the if direction we proceed by induction on the size of the graph. The statement is clear for graphs with 2 or fewer vertices. Now suppose you have a graph G where $|G| > 2$ which is not a join in any colour. Let $x \in G$, note that $G \setminus \{x\}$ can be written as a coloured join of non-empty graphs by assumption, and for convenience call this colour **blue** or b . Let y, z be vertices in distinct **blue**-disconnected components. Since G is **blue**-disconnected and $p_b(G)^\vee$ is a cograph, we know that $y - x - z$ are connected via **cobblue** colours. By the triangle assumption, $x -_r y -_r z$ for some other colour r we call **red**. By the same argument, if w is any other vertex **blue**-disconnected from y then $x -_r w$. Now let $a - y$ via a **cobblue** colour. Since **cobblue** is a cograph and $a - y - x - z$, we must have $a -_r x$. By iterating, we conclude any vertex in the **blue**-coconnected component of y is **red**-connected to x . But then G is a **red**-join, which is a contradiction. $\square$

Remark 2.1.16. Proposition 2.1.15 allows us to give explicit descriptions of coloured versions of $\mathbb{E}$ and $\mathbb{D}$. In particular, letting P be a poset and $x \in P$, let $\mathbb{E}_{P,x}$ be the smallest subcategory of Graph_P containing the x -reflexive 1-vertex graph and closed under finite coloured joins. The categories $\mathbb{E}_{P,x}$ have universal properties analogous to Proposition 2.1.10, which we claim are universal both as 1-categories and ∞ -categories.

Proposition 2.1.17. *Let $P = 1 < \dots < n$ be a finite totally ordered poset. Then the category $\mathbb{E}_{P,i}$ is an n -fold monoidal category in the sense of [BFSV98], and is the free n -fold monoidal category on a monoid for the i^{th} monoidal product.*

2.2 Twofold monoidal categories

Note that $\mathbb{E}$ is a perfect operator category. We can identify its Leinster category $\mathbb{E}_* := \Lambda(E)$ with the category of reflexive cographs and partial maps, where a partial map from G to H is an induced subgraph $G' \hookrightarrow G$ and a graph morphism $f : G' \rightarrow H^1$. The induced inert-active factorisation on $\mathbb{E}_*$ can be described as follows: a partial graph morphism $(G' \subset G, f : G' \rightarrow H)$ is inert if f is a graph isomorphism. It is active if $G' = G$.

Let $\delta, \tau : F \rightarrow E$ be the functors that identify a finite set X with, respectively the completely disconnected and the complete cograph on $|X|$ vertices. We will also denote by δ, τ the induced functors $F_* \rightarrow E_*$. In upcoming work, Barwick gives the following definition of a 2-fold monoidal category [Bar24]:

Definition 2.2.1. A **2-fold symmetric monoidal category** is a functor $p : C^{\oplus, \otimes} \rightarrow E$ satisfying the following conditions:

1. p is a locally coCartesian fibration.

¹Equivalently, we can see it as the category of pointed reflexive cographs and pointed graph morphisms, where a pointing on a reflexive cograph is a choice of vertex connected to every other vertex by an edge, and a pointed graph morphism is a graph morphism that preserves the specified vertex.

continue
from
here

2. The locally p -coCartesian edges over inert edges are p -coCartesian,
3. For each $X \in C^{\oplus, \otimes}$ over a cograph G , the locally p -coCartesian lifts of the inert morphisms $(p_i : G \rightarrow 1)_{i \in |G|}$ assemble to an equivalence $X \simeq \prod_{i \in G} X_i$ where $p_i^* : X \rightarrow X_i$ is the locally p -coCartesian lift of p_i ,
4. The pullback of p along δ and τ are both symmetric monoidal categories.

A **twofold symmetric monoidal functor** of 2-fold monoidal categories is a morphism in $Cat_{/\mathbb{E}_*}$ between twofold symmetric monoidal categories that preserves locally coCartesian edges. For such morphisms $F : C^{\oplus, \oplus} \rightarrow D^{\oplus', \oplus'}$ we automatically have that the induced morphisms $C^{\oplus} \rightarrow D^{\oplus'}$ and $C^{\oplus} \rightarrow D^{\oplus'}$ is symmetric monoidal. The category of twofold symmetric monoidal category $2SMonCat$ is the subcategory of $Cat_{/\mathbb{E}_*}$ consisting of twofold symmetric monoidal categories and 2-fold symmetric monoidal functors between them.

Remark 2.2.2. The terminal 2-fold monoidal category is $id : E_* \rightarrow E_*$.

Remark 2.2.3. We would now like to give a twofold version of the comonad $U'Env'$ of Theorem 1.3.4. It is clear to us that this should be a comonad on the category $2SymMon_{/\mathbb{E}^{\oplus, \oplus}}$. This is because we have an equivalence $Act(\mathbb{E}_*) \simeq \mathbb{E}^{\oplus, \oplus}$. We can certainly let $Act(E_*)$ be the full subcategory of $Fun(\Delta^1, E_*)$ spanned by active morphisms, let $s, t : Act(E_*) \rightarrow E_*$ be the functors corresponding to source and target, and form the pull-back

$$\begin{array}{ccc} Env(C)^{\oplus, \oplus} & \longrightarrow & C^{\oplus, \oplus} \\ \downarrow & & \downarrow \pi \\ Act(\mathbb{E}_*) & \xrightarrow{s} & \mathbb{E}_* \end{array}$$

with the functor to $\mathbb{E}_*$ given by t . However, `crefrmk:cocartesian-lift` suggests that this will be a coCartesian fibration, since the coCartesian lifts only used the inert-active factorisation system and the coCartesian lifts of inerts. For this reason, we are unsure how to proceed. A major goal of the project is to give a working envelope construction, or alternatively, a direct description of the desired subcategory of $2SymMon_{/\mathbb{E}^{\oplus, \oplus}}$ a la [HK22][2.4.6.].

2.3 The category $\mathbb{H}$

The purpose of this section is to suggest an answer to the following question:

Question 2.3.1. *What is the free twofold monoidal category on an object that is both a $\oplus$ -monoid and a $\oplus$ -comonoid?*

Note that we are **not** asking for these structures to be compatible with the interchange map. We are interested in this construction, as we expect certain fibrations over it to give the category $2Op$ of twofold operads as a generalisation of the compound pseudo-tensor categories of Beilinson and Drinfeld [BD04].

In particular, we expect it to encode the data of an operad, a cooperad, and compatibilities between them.

Recall the following construction.

Definition 2.3.2. Suppose C is a 1-category² with all pullbacks. Then the **category of spans in C** , $Span(C)$ has the same objects as C and a morphism from x to y is given by an object $z \in C$ and morphisms

$$\begin{array}{ccc} & z & \\ f \swarrow & & \searrow g \\ x & & y \end{array}$$

in C , with composition given by pullback.

There is a faithful embedding $\iota : C \rightarrow Span(C)$ given by sending $f : x \rightarrow y$ to the span

$$\begin{array}{ccc} & x & \\ \parallel \swarrow & & \searrow f \\ x & & y \end{array}$$

Similarly, there is a faithful embedding $\iota' : C^{op} \rightarrow Span(C)$ that sends $f^{op} : y \rightarrow x$ to the span

$$\begin{array}{ccc} & x & \\ f \swarrow & & \searrow \parallel \\ y & & x \end{array}$$

If furthermore C is equipped with a twofold monoidal structure $(C, \underline{\oplus}, \overline{\oplus}, 1)$, then C^{op} can canonically be equipped with the twofold monoidal structure $(C^{op}, \overline{\oplus}, \underline{\oplus}, 1)$. In this case, there are two canonical twofold monoidal structures on $Span(C)$ for which respectively ι and ι' are strong twofold monoidal. We refer to $(Span(C), \underline{\oplus}, \overline{\oplus}, 1)$ for the former, and $(Span(C), \overline{\oplus}, \underline{\oplus}, 1)$ for the latter.

We will use this property of the span construction to our advantage. Recall that $\mathbb{D}$ is the free twofold monoidal category on a monoid for the left product. We consider it a subcategory of $Graph^{ir}$, the twofold category of irreflexive graphs, which we note has all pull-backs. Indeed, it has pull-backs over 1, and so the general pull-back has vertices the disjoint union of pull-backs over the fibers, with the corresponding maps, and maximal edges such that these maps are graph maps. By the universal property of D , $Span(Graph^{ir})$ contains both a free $\underline{\oplus}$ -monoid and a free $\overline{\oplus}$ -comonoid. These can be identified with $F \cong Span(F, F^{id}, F) \rightarrow Span(Graph^{ir})$ sending a set to the discrete graph on that many vertices, and $F^{op} \cong Span(F, F, F^{id}) \rightarrow Span(Graph^{ir})$ sending a function $f^{op} : Y \rightarrow X$ to the span

$$\begin{array}{ccc} & disc(X) & \\ f \swarrow & & \searrow \\ comp(Y) & & comp(X) \end{array}$$

²This definition can be extended to ∞ -categories with pullbacks, but we won't need this.

where $comp(X)$ is the complete graph on $|X|$ vertices, $disc(X)$ is the discrete graph on $|X|$ vertices, and the graph maps are induced by their functions on vertices, where the right map is induced by the identity on X . We note that these form subcategories, and lazily denote these $F, F^{op} \subset Span(Graph^{ir})$.

Let $\mathbb{H}$ be the smallest sub-twofold monoidal category of $Span(Graph^{ir})$ containing F and F^{op} and closed under $\oplus$ and $\overline{\oplus}$. By a sub-twofold monoidal category we mean a strong two-fold monoidal functor that is faithful and injective on objects.

Remark 2.3.3. Let us make the following remarks about $\mathbb{H}$

- The map $\mathbb{H} \rightarrow Span(Graph^{ir})$ factors through the map $Span(\mathbb{D}, D^{acc}, D) \rightarrow Span(Graph^{ir})$.
- Every morphism in $\mathbb{H}$ has the following property: every fiber of the backwards map is sent dispersively to a complete graph by the right map.

Let $Span^*(\mathbb{D})$ be the category of spans of irreflexive cographs whose backwards maps are accretive and such that the above fiber condition is satisfied.

Remark 2.3.4. Let us remark on why the two categories mentioned above exist. Firstly, $Span(Graph^{ir})$ exists because $Graph^{ir}$ has pullbacks. Indeed, the pullback of $f : X \rightarrow A$ along $g : Y \rightarrow A$ has as set of vertices the pullback $|X| \times_{|A|} |Y|$ taken in Set , which also gives the maps f', g' on vertices. Then the universal property is exactly satisfied if two vertices $x, y \in X \times_A Y$ are joined by an edge iff $f'(x), f'(y)$ are joined by an edge and $g'(x), g'(y)$ are joined by an edge.

To see why $Span^*(\mathbb{D})$ is a subcategory of this, consider first that since accretives pull back to accretives, the property that the backwards map is accretive is preserved by composition. Similarly, the fiber condition is satisfied. Finally, we should argue that the composition of two such spans of irreflexive cographs has a cograph apex. To see this, let $f : B \rightarrow X$ and $a : A \rightarrow X$ be maps of cographs such that a is accretive, and take their pullback $A \times_X B$ in $Graph^{ir}$ as below.

$$\begin{array}{ccc} A \times_X B & \xrightarrow{f'} & A \\ a' \downarrow & \lrcorner & \downarrow a \\ B & \xrightarrow{f} & X \end{array}$$

Now suppose for sake of contradiction that N is an induced subgraph of $A \times_X B$. Since a' is accretive, it cannot be dispersive. However, any morphism out of N is either dispersive or not accretive, which is a contradiction.

Proposition 2.3.5. $\mathbb{H}$ is equivalent to the category of spans of irreflexive cographs whose backwards maps are accretive, and such that each fiber of the backwards map is sent to a complete graph by the forwards map.

2.4 Next steps

This section will record questions and next steps for this project. Everything said here is speculative and unapologetically vague.

- I. A major goal is to compare the twofold monoidal categories of [Bar24] with the duoidal categories of [Tor24]. In particular, the category of twofold monoidal categories should be equivalent to the category of duoidal categories where the units are equivalent. This would additionally allow us to confirm the proposed universal properties of $\mathbb{D}$, $\mathbb{E}$ and $\mathbb{H}$.
- II. A non-symmetric version of twofold monoidal categories exists, using certain directed graphs. We say a $[n, m]$ -category is a twofold category that is an E_n -monoidal category with respect to the left tensor product and E_m -monoidal category with respect to the right tensor product. In upcoming work with Clark Barwick, we will show that the functor $SMonCat \rightarrow [\infty, n] - MonCat$ assigning to a symmetric monoidal category $C^\otimes$ the repeated $[\infty, n]$ -monoidal structure $C^{\otimes, \otimes}$ has a left adjoint given by inverting the interchange map, and explore related results.
- III. As it stands, the description of $\mathbb{H}$ is somewhat esoteric. We would like to find a more natural description whose morphisms are easier to describe.
- IV. We believe certain fibrations over $\mathbb{H}$ can give another description of twofold symmetric monoidal categories, and we would like to give this description. It would serve as an intermediate category between $\mathbb{F}^{op} \times \mathbb{F}$ and $\mathbb{E}$, and could be compared to duoidal categories and twofold monoidal categories in the sense of [Bar24]. In particular, since $\mathbb{H}$ is inherently "twisted", the duoidal structure on a twofold monoidal category should be pulled back along a natural map (or possibly maps) $\mathbb{F}^{op} \times F \rightarrow \mathbb{H}$. Similarly, it should be simple to identify a compound tensor category with its corresponding twofold symmetric monoidal category in this framework.
- V. Following on from the last point, we are interested in giving a definition of twofold operads. Ideally these would be certain fibrations over $\mathbb{H}$ which support a canonical inclusion of twofold monoidal categories to twofold operads. As an alternative or intermediate step, the methods of [HK22] can be used to define twofold operads as certain twofold symmetric monoidal categories equipped with a twofold monoidal functor to $\mathbb{H}$.
- VI. We are interested in twofold operads mainly because we believe the theory has interesting applications, which we would like to explore. For one, it will be possible to generalise the results in [BD04] to the ∞ -categorical world. It is also our hope that the self-duality of twofold operads can give a natural description of Koszul duality. A major goal of the project going forward will be to understand Koszul duality for operads to understand if they can be described using twofold operads.

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